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System and method of detecting a welded contact

Abstract

An industrial switching device prevents a false indication of an open electrical circuit when contacts within the switching device are welded shut. Under normal operation, an actuator is selectively moved between a first position and a second position to reciprocally move a plunger to open and close the contacts. The actuator includes an indicator of whether the switching device is in the “ON” or “OFF” state. When the contacts are welded shut, a plunger engagement section interlocks with an actuator engagement section to prevent the actuator from moving from the “ON” position to the “OFF” position. A crossbar on the plunger engagement section embeds in the actuator engagement section to prevent the actuator from sliding over the plunger. A side member on the plunger engagement section prevents the actuator from sliding around the plunger.

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Background/Summary

BACKGROUND INFORMATION

(1) The subject matter disclosed herein relates to switching devices for industrial equipment. More specifically, an aspect of the invention relates to switching devices that include a mechanism and method of preventing undesired movement of the actuator indicating that the electrical connection of the switching device is open when the electrical connection is still closed.

(2) As is known to those skilled in the art, switching devices are components in an electrical circuit that may be controlled between an “ON” and an “OFF” state. In the ON state, the switching device establishes an electrical connection between contacts and allows electrical current to flow through the switching device from a power source to an electrical load. In the OFF state, the switching device opens, or breaks, the electrical connection between contacts, preventing the electrical current from flowing through the switching device. Switching devices may be used, for example, as a circuit breaker, motor protection device, contactor to supply power to one or more branch circuits, or the like. The switching device may be manually or automatically actuated. A mechanical switch or electronic actuator is provided which moves between two states. In one state, the mechanical switch or electronic actuator causes the switching device to establish the electrical connection, and, in the other state, the mechanical switch or electronic actuator causes the switching device to break the electrical connection. The mechanical switch may be actuated by

rotating a handle of a rotary disk, which causes a plunger to move a switching element within the switching device in a first direction as the switch or actuator transition from a first state to a second state. When the mechanical switch or electrical actuator, transition back from the second state to the first state, a spring causes the switching element to return to its original position.

(3) However, these switching devices are not without certain challenges. In certain failure modes, the potential exists for the contacts to be welded together. For example, a short circuit in a motor either between windings or to ground due to a failure in the insulation within the motor may result in a significant amount of current drawn thorough the motor. While the branch circuit to the motor includes a circuit breaker configured to trip upon detection of such an inrush of current, the amplitude of current through the contacts of the electrical switch device may be high enough for a short duration to weld the contacts together. When the contacts are welded together, the plunger is unable to move. The switching device therefore remains in the first state, where the electrical connection remains established and cannot be broken.

(4) Additionally, the potential further exists for the switch to appear in an OFF state, indicating the electrical connection is open, even when it is not. The actuator may include, or the handle of the actuator may serve as, an indicator pointing to indicia on the switch identifying whether the switch is ON or OFF. A technician may turn the actuator from an ON state to an OFF state in advance of maintenance on the industrial equipment connected to the switch. When the technician rotates the actuator, “slippage” occurs within the switch, allowing components within the switch to slide over one another such that the actuator appears to be in the OFF state even when the contacts are welded together. As a result, the orientation of the actuator does not indicate the true state of the electrical connection. The actuator appears to indicate the electrical connection switch is OFF when it is not, creating a dangerous situation.

(5) While attempts have been made to solve such an issue, such attempts have not fully solved the issue without incurring various certain disadvantages. Thus, it would be desirable to provide a new mechanism that does not allow the actuator to “slip” and incorrectly indicate that the switch is in an OFF state. Further, it would be advantageous if the new mechanism was relatively easy and inexpensive to implement.

BRIEF DESCRIPTION

(6) According to one embodiment of the invention, a switching device for industrial equipment, having at least one pair of contacts, includes an actuator and a plunger. The actuator has an actuator engagement section, and the actuator is selectively moved between a first position and a second position. The plunger has an engagement area and a crossbar. When the at least one pair of contacts are not welded together, the actuator selectively moves the plunger as the actuator is moved between the first position and the second position. When the at least one pair of contacts are welded together, the actuator engagement section engages the crossbar thereby preventing the actuator from moving between the first state and the second state.

(7) According to another embodiment of the invention, a method of detecting contacts welded together in an industrial switching device is disclosed. An actuator of a switching device for industrial equipment is attempted to be rotated from an “ON” state towards an “OFF” state. An actuator engagement section of the actuator engages with a plunger engagement section of a plunger while attempting to rotate the actuator when the contacts are welded together, and the actuator selectively moves the plunger between a first position and a second position when the contacts are not welded together. The actuator is prevented from rotating from the “ON” state to the “OFF” state as a result of engaging the actuator engagement section with the plunger engagement section when the contacts are welded together, and the contacts are detected as being welded together as a result of preventing the actuator from rotating.

(8) According to still another embodiment of the invention, a switching device for industrial equipment, having at least one pair of contacts, includes a plunger having an engagement area. The engagement area includes a long member extending across a direction of rotation of an actuator for

the switching device and a short member extending tangential to a direction of rotation. The long member is configured to prevent rotation of the actuator when the at least one pair of contacts are welded together, and the short member is configured to prevent the actuator from moving laterally around the plunger. The engagement area includes a crossbar configured to engage an actuator engagement section of the actuator to prevent the actuator from moving over the plunger.

(9) These and other advantages and features of the invention will become apparent to those skilled in the art from the detailed description and the accompanying drawings. It should be understood, however, that the detailed description and accompanying drawings, while indicating preferred embodiments of the present invention, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the present invention without departing from the spirit thereof, and the invention includes all such modifications.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various exemplary embodiments of the subject matter disclosed herein are illustrated in the accompanying drawings in which like reference numerals represent like parts throughout, and in which:

(2) FIG. 1 is a schematic diagram of a series of electrical switches connected to a fuse;
(3) FIG. 2 is a front elevation view of one of the electrical switches of FIG. 1;
(4) FIG. 3 is a side elevation view of the electrical switch of FIGS. 1 and 2 with a lower housing removed;

(5) FIG. 4 is a partial perspective view of a switching device;

(6) FIG. 5 is a partial front elevation view of the electrical switch of FIGS. 1-3;

(7) FIG. 6 is a perspective view of one embodiment of an actuator;

(8) FIG. 7 is a side elevation view of the actuator of FIG. 6;

(9) FIG. 8 is a perspective view of one embodiment of a plunger;

(10) FIG. 9 is an elevation view of the plunger of FIG. 8;

(11) FIG. 10 is a partial perspective view of the plunger of FIGS. 8 and 9;

(12) FIG. 11 is a right perspective view of the actuator of FIG. 6 and the plunger of FIGS. 8-10;

(13) FIG. 12 is a left perspective view of the actuator and the plunger of FIG. 11;

(14) FIG. 13 is an elevation view of the actuator and the plunger of FIGS. 11 and 12;

(15) FIG. 14A is a partial sectional view of another embodiment of the actuator and plunger;

(16) FIG. 14B is a partial sectional view of yet another embodiment of the actuator and plunger;

(17) FIG. 15 is a partial section view of the actuator and an internal mechanism to move the plunger with the switching device; and

(18) FIG. 16 is a partial perspective view of the actuator of FIG. 6 and the plunger of FIG. 9 incorporated in a switching device.

(19) In describing the various embodiments of the invention which are illustrated in the drawings, specific terminology will be resorted to for the sake of clarity. However, it is not intended that the invention be limited to the specific terms so selected and it is understood that each specific term includes all technical equivalents which operate in a similar manner to accomplish a similar purpose. For example, the word "connected," "attached," or terms similar thereto are often used. They are not limited to direct connection but include connection through other elements where such connection is recognized as being equivalent by those skilled in the art.

DETAILED DESCRIPTION

(20) The various features and advantageous details of the subject matter disclosed herein are explained more fully with reference to the non-limiting embodiments described in detail in the following description.

(21) The subject matter disclosed herein describes a system of preventing electrical switch devices for industrial equipment from providing an indication that the switch is in an OFF state when it is not. Switching devices may include an actuator which, when actuated, allows the switching device to alternate between a first state, where there is an electrical connection, and a second state, where the electrical connection is broken. The switching device alternates between the two states by selectively closing or opening contacts within the switching device. The switching device includes an actuator with an actuator engagement section. The switching device further includes a plunger with a plunger engagement section. The plunger engagement section has an engagement area. At least a portion of the actuator engagement section is selectively rotatable toward the engagement area of the plunger. When the contacts are not welded, rotation of the actuator causes the plunger to move into the switching device and away from the actuator allowing the actuator engagement section to pass by the plunger engagement section. When the contacts are welded, the welded contacts prevent the plunger from moving into the switching device during rotation of the actuator and the actuator engagement section engages the plunger engagement section. The plunger engagement section is configured to stop both rotation of the actuator and lateral motion of the actuator to prevent the actuator from slipping besides the plunger.

(22) In another embodiment of the invention, the invention may further include a switching device that has an actuator with an actuator sidewall. The switching device further includes a plunger with a plunger engagement section. The plunger engagement section further has a crossbar. The actuator sidewall is formed from a first material and the crossbar is formed from a second material. The first material is softer than the second material. When the contacts are welded and the actuator is rotated, the actuator sidewall and crossbar interlock. The crossbar engaging with the actuator sidewall is configured to prevent the actuator from slipping over the plunger.

(23) The novel switching device therefore prevents the actuator from slipping besides or over the plunger when the contacts are welded and falsely indicating that the electrical connection has been broken.

(24) Turning initially to FIG. 1, electrical switches 5 may be mounted in a cabinet and used to control industrial equipment. Group installation allows multiple motors or loads to be connected to a single branch circuit protection device. A separate electrical switch 5 is mounted between the branch circuit protection device and each motor or load. As illustrated in FIG. 2, an electrical switch 5 includes a housing 10 with a housing aperture 15. The electric switch 5 includes an actuator 20, that extends through the housing aperture 15, and a switching device 25 (shown in FIG. 4) that may be incorporated into the electric switch 5. The electric switch 5 may be a circuit breaker, a motor protection circuit, an electrical contactor, or the like. The switching device 25 includes a plunger 30 configured to move reciprocally, back-and-forth, along an axis 35. A first end 32 of the plunger 30 is configured to engage the actuator 20 in the electrical switch 5. It is contemplated that the plunger 30 may directly engage the actuator 20 or include one or more intermediate members within the switch 5 to cause the reciprocal motion as the actuator 20 moves between two positions. The actuator 20 may be manually activated by, for example, a toggle switch or a handle 24 on a rotary switch 22 (see e.g., FIG. 2). Optionally, the actuator 20 may be automatically activated, for example, by a solenoid energized by a relay. Regardless of manual or automatic actuation, the plunger 30 is configured to move between a first position and a second position. The second end of the plunger 30 (not shown) is configured to be inserted into a plunger seat (not shown). The plunger seat may have a lower surface against which the second end of the plunger 30 rests. Optionally, the plunger seat may have an open end and the plunger seat is configured to align the plunger 30 with the switching element 25. The second end of the plunger 30 may then rest against the switching element 45. The plunger 30 applies a force to the switching element 45 when the actuator 20 drives the plunger 30 from the first position to the second position either directly, if the second end of the plunger 20 rests on the switching element 45, or indirectly via the plunger seat if the plunger seat includes a lower surface. The illustrated plunger 30 is

intended to be exemplary only. It is illustrated as a unitary cylindrical member being driven reciprocally along the axis of the piston. However, it is contemplated that multiple plungers **30** may be mechanically connected or formed as a single member such that multiple switching elements **45** within a switching device **25** may be selectively opened or closed in tandem. According to one aspect of the invention, it is contemplated that the switching device **25** may include three plungers **30** each driven in tandem to selectively open or close a three phase electrical connection. It is further contemplated that a single plunger **70** may include multiple arms **170**, each configured to engage a separate switching element within the switch (see e.g., FIG. **9**). It is further contemplated that the geometry of the plunger **30** may take other forms, such as a square or rectangular sectional area. It is also contemplated that the plunger **30** may include an offset segment along the length of the plunger such that a force is applied at a first end of the plunger **30** along a first axis and the second end of the plunger **30** moves reciprocally along a second axis where the second axis is parallel to but offset from the first axis. The offset segment may, for example, run generally orthogonal between two additional members, where the first additional member extends generally along the first axis and the second additional member extends generally along the second axis.

(25) Terms such as upper, lower, inner, outer, front, rear, left, right, and the like will be used herein with respect to the illustrated switching device **25**. These terms are relational with respect to the illustrated switching device and are not intended to be limiting. It is understood that the switching device **25** may be installed in different orientations, such as vertical or horizontal, or may be rotated one hundred eighty degrees without deviating from the scope of the invention.

(26) With reference also to FIG. **4**, the switching device **25** includes a spring **50** which applies a force in a direction opposing actuation of the plunger **30**. The spring **50** has a first end which is seated against a lower housing of the switching device **25** and a second end which is seated, at least in part, against a lower surface **31** of the switching element **45**. When the plunger **30** is in the first position, the spring **50** may be partially compressed such that it applies a sufficient force against the switching element **45** to stay in the first position. As the actuator drives the plunger **30** from the first position to the second position, the spring **50** becomes more compressed, applying a greater force against the switching element **45**. When the actuator **20** releases the plunger **30**, the spring **50** causes the switching element **45** to return to and stay in the first position. If multiple switching elements **45** exist within a switch, each switching element **45** may have a spring associated with the element to exert a force on the plunger. According to still another embodiment, a plunger **70** may be configured to engage multiple switching elements **45** in tandem and the plunger **70** may have a spring seat **159** against which a single spring may be positioned, applying a biasing force directly against the plunger **70** rather than against one or more switching elements **45**.

(27) The switching element **45** is an assembly that includes at least one contact **65** mounted on the switching element **45**. The contact **65** is configured to engage a complementary contact **67** in the first position and to disengage from the complementary contact **67** as the switching element **45** is moved from the first position to the second position. According to the illustrated embodiment, the switching element **45** includes a pair of contacts **65**, where one contact is mounted to the switching element on one side of the plunger **30** when the plunger engages the switching element and the other contact is mounted on the other side of the plunger **30**.

(28) With reference again to FIGS. **2-4** and stated previously, the electric switch **5** has an actuator **20** which engages and drives the plunger **30** from a first to second position thereby causing the switching device **5** to establish the electrical connection in a first state through connection between the movable contacts **65** and the fixed contacts **67** or to break the electrical connection between the contacts **65**, **67** in a second state. The movable contacts **65**, however, may become welded to the fixed contacts **67** when the switch **5** is overloaded or experiences a brief, very high current due, for example, to a short circuit in the load connected to the switch. When the movable contacts **65** are welded to the fixed contacts **67**, the force applied from the actuator **20** is insufficient to separate the contacts. Further, if the actuator **20** were able to rotate despite the welded contacts, the handle **24** of

the actuator could indicate that the switch is in a second state (i.e., the electrical connection has been broken) even when the welded contacts maintain the electrical connection. The switching element **45** is illustrated in FIG. **3** in the first state with the movable contacts **65** on the switching device engaging the fixed contacts **67** on the upper portion of the switch **5**. In this position, an electrical connection is established, allowing current to flow through the switch **5**.

(29) In one embodiment, the actuator **20** may be a rotary disk. Prior actuators had the potential for “slipping” when actuated. In other words, when the prior actuator was actuated through rotation, the prior actuator had the potential to slip over or beside a plunger so that the actuator indicated the switching device **25** had moved to the second state, or OFF condition. However, if the movable contacts **65** on the switching device **25** and the complementary contacts **67** were welded, the electrical connection remained established and only appeared, via the orientation of the actuator, to have been broken.

(30) With reference to FIGS. **6-7**, one embodiment of the invention includes an actuator **20** configured for rotational motion. In the illustrated embodiment, the actuator **20** may include a first actuator face **75** and a second actuator face **80**, located and positioned opposite from the first actuator face **75**. The actuator **20** further includes three segments defined between the second actuator face **80** and the first actuator face **75**. As best illustrated in FIG. **7**, a first segment defines an actuator engagement section **85** located and positioned proximate the second actuator face **80** and extending for a first distance from the second actuator face **80** toward the first actuator face **75**. A second segment defines an actuator shoulder section **90** beginning at the actuator engagement section **85** and extending away from the actuator engagement section **85** and towards the first actuator face **75** for a second distance. The third segment defines an actuator handle section **95** beginning at the actuator shoulder section **90** and further projecting away from the actuator shoulder section **90** for a third distance to the first actuator face **75**.

(31) The actuator engagement section **85** extends below and along one side of the actuator **20**. A lower surface of the actuator engagement section **85** generally defines the second actuator face **80**. The actuator engagement section **85** includes an actuator engagement sidewall **125** extending generally orthogonally away from the second actuator face **80** for at least a portion of the first distance. According to the illustrated embodiment, the actuator engagement sidewall **125** has a rounded edge joining the second actuator face **80** and extends the rest of the first distance to the actuator shoulder section **90**. The outer periphery has a generally arcuate shape extending between a first end **132** and a second end **134** of the actuator engagement section **85** with a protrusion **133** extending axially outward from the actuator engagement sidewall part way between the first and second ends. An actuator engagement surface **130** is located proximate the second end **134** of the actuator engagement section **85** and is generally “U”-shaped. The inner periphery of the actuator engagement section **85** is a generally planar surface returning from the second end **134** to the first end **132** of the actuator engagement section below the actuator shoulder section **90**.

(32) The actuator shoulder section **90** defines a middle portion of the actuator **20** and is configured to slidably engage other components within the switch **5** as the actuator **20** is rotated between a first position and a second position. The actuator shoulder section **90** has an irregular geometric outer periphery with a first portion **91** of the outer periphery being generally semicircular and a second portion **93** of the outer periphery including a cam **97** extending therefrom. The cam **97** is configured to engage either a first gear tooth **141** or a second gear tooth **143** (see FIG. **11**) to cause the reciprocal motion of the plungers in the switching element **45**. When the actuator **20** is rotated in a counter-clockwise motion, the cam **97** engages the first gear tooth **141**, rotating the internal actuator **147** from a first position to a second position. When the actuator **20** is rotated in a clockwise motion, the cam **97** engages the second gear tooth **143**, rotating the internal actuator **147** from the second position back to the first position. The internal actuator **147** includes a second cam which, in turn, engages a plate **149**, coupled between the internal actuator **147** and the plunger **70**, to create linear motion of the plate **149**. The linear motion of the plate **149** is reciprocal away from

and returning toward the actuator **20**. Because the plunger **70** is coupled to the plate **149**, the plunger **70** similarly moves in a reciprocal motion away from and returning toward the actuator **20** as the actuator **20** is rotated between the first and second positions.

(33) The actuator handle section **95** is positioned at the front most surface of the actuator **20**. The actuator handle section **95** may be shaped and sized so that the actuator handle section **95** extends through the housing aperture **15** of the switch **5** and is configured to receive a handle **22** within the channel **101** on the front of the actuator **20**. The actuator handle **22** may further include a generally circular actuator handle face **23** (see also FIG. 2) and a generally rectangular grip **24** extending perpendicular to the actuator handle face. The grip **24** is shaped and sized so that a user may grasp the grip to rotate the actuator **20**. In one aspect of the invention, the grip **24** may additionally provide an indication of whether the switch **5** is in the first state or the second state by “pointing” to “on” and “off” labels on the housing **10** of the switch **5**.

(34) In an aspect of the invention, at least the actuator engagement surface **130** of the actuator **20** may be formed from a first material. The first material may be nylon, polyoxymethylene (POM), polybutylene terephthalate (PBT), or any combination of such material. The first material is preferably not reinforced with glass fiber or is reinforced with a low amount of glass fiber, or any other suitable material which is a softer material than the material forming at least a portion of the plunger **70**, as will be described hereinafter. Typically, the entire actuator **20** will be formed as a single element and made from a common material.

(35) As shown in FIGS. 8-10, in one aspect of the invention, the plunger **70** includes a first plunger face **135** and a second plunger face **140** located opposite from each other. The plunger **70** further includes a first plunger end **145** and a second plunger end **150**, located opposite from the first plunger end **145**, as well as a first plunger side **155** and a second plunger side **160** also located opposite from each other. An outer periphery, also referred to herein as a plunger sidewall **165**, is defined by the first plunger end **145**, which meets the first plunger side **155**, which, in turn, meets the second plunger end **150**, and which, in turn meets the second plunger side **160** moving around the plunger. The plunger sidewall **165** extends generally perpendicular between both the first plunger face **135** and the second plunger face **140**.

(36) As illustrated, the plunger **70** further includes three plunger arms **170**, with each plunger arm **170** generally being a Z-like member. More specifically, each plunger arm **170** may have three regions, a first plunger arm region **170A**, a second plunger arm region **170B**, and a third plunger arm region **170C**. The first plunger arm region **170A** projects away from the first plunger sidewall **165** at the plunger side **155** before extending into the second plunger arm region **170B**. The second plunger arm region **170B** projects away from the second plunger face **140** and generally perpendicular to the first plunger arm region **170A**. The plunger arm **170** then continues to extend, where the third plunger arm **170C** projects generally perpendicular to the second plunger arm region **170B**, away from the first plunger side **155**. The plunger arm regions **170B** is therefore located between the first and third plunger arm regions **170A** and **170C**, creating an offset between the first and third plunger arm regions. The first and third plunger arm regions **170A** and **170C** extend generally parallel to but in different axes from one another to form the Z-like member. Each arm **170** preferably engages a switching element **45**, thereby causing movable contacts **65** to selectively separate from or connect with the fixed contacts **67** to enable or disable an electrical current. In other embodiments, the plunger **70** may include a greater or lesser number of arms.

(37) The plunger **70** further includes a plunger member **172**. The plunger member **172** extends between a first end **171** and a second end **173** and includes a first plunger member region **172A**, a second plunger member region **172B**, and a third plunger member region **172C**. In one embodiment, the first plunger member region **172A** begins at the first end **171** of the plunger member **172** and projects away from the second plunger side **160** in the direction opposite from the first plunger arm region **170A**, the first plunger member region **172A** extends to the second plunger member region **172B**, which is closer to the second end **173** than the first end **171** of the plunger

member **172**. The second plunger member **172B** has a width less than the width of the first plunger member **172A** and extends in a generally orthogonal direction to the first plunger member region **172A** along a portion the end of the first plunger member region **172A** that is distal from the second plunger side **160**. The third plunger member region **172C** also has a width less than the width of the first plunger member **172A** and is positioned adjacent to the second plunger member **172** along the end of the first plunger member region **172A** that is distal from the second plunger side **160**. The third plunger member region **172C** extends from the first plunger member region **172A** in a direction that is generally coplanar to the first plunger member **172A**. The third plunger member regions **172** will also be referred to herein as the plunger engagement section **175**.

(38) In one embodiment, the plunger engagement section **175** extends from a first surface **176** to a second surface **177**, the second surface located opposite the first surface **176**. The first surface **176** may, in part, be integrally molded with the first plunger member region **172A**, and the second surface **177** defines the second end **173** of the plunger member **172**. The plunger engagement section **175** is additionally L-shaped extending between a first end **179** and a second end **181**. The first end **179** of the plunger engagement section **175** defines the top of the L-shape, and the second end **181** of the plunger engagement section **175** defines the long portion of the L-shape. The long portion of the L-shape is oriented transverse to the direction of rotation of the actuator and is configured to prevent rotation of the actuator **20** when the contacts are welded. The short portion of the L-shape extends generally orthogonal to the long portion and tangential to the direction of rotation of the actuator. The short portion of the L-shape is configured to extend towards the direction of the ON position such that the actuator engagement section **85** rotates within the long and short portions of the L-shape as it is rotated from the ON position to the OFF position. The short portion of the L-shape is configured to prevent the actuator **20** from sliding laterally around the plunger **70**. The plunger engagement section **175** may also include a plunger engagement sidewall **178** extending downwards from the second surface **177**. The plunger engagement sidewall **178** forms an indentation or engagement area **190** located between the second surface **177** of the plunger engagement section **175** and the first plunger member region **172A**. The plunger engagement section **175** further includes a crossbar **195** located adjacent to the edge of the second surface **177** of the plunger engagement section **175** and protruding from the plunger engagement sidewall **178**. The crossbar **195** may be shaped like a triangle, blade, or other angled protrusion so that a base of the crossbar **195** is located adjacent to the plunger engagement sidewall **178** and narrows into a point when viewed cross sectionally.

(39) At least a portion of the plunger engagement section **175**, and preferably the crossbar **195**, may be formed from a second material. The second material may be polyether ether ketone (PEEK) and reinforced with carbon fiber, glass reinforced polyphenylene sulfide (PPS GF), polyamides with glass particles or nylon with glass fiber (PA GF), or any other material of sufficient hardness, as will be explained hereinafter. Typically, the entire plunger **70** will be formed as a single element and made from a common material.

(40) Referring to FIGS. **11-13**, the plunger engagement section **175** selectively engages with the actuator engagement section **85** when at least one set of the mobile contacts **65** are welded to the stationary contacts **67** and an attempt is made to rotate the actuator **20**. In more detail, as the actuator **20** is partially turned from the ON state and the actuator engagement section **85** rotates into the engagement area **190** of the plunger engagement section **175** until the actuator engagement surface **134** contacts the crossbar **195** of the plunger **70**. The L-shaped engagement area **190** of the plunger **70** prevents the actuator **20** from slipping besides the plunger **70**, while the crossbar **195** prevents the actuator **20** from slipping over the plunger **70**. The harder material of the crossbar **195** is configured to cause the crossbar **195** to interlock with the softer material of the actuator engagement surface **130**. The crossbar **195** is able to “bite”, or compress, into the softer material of the actuator engagement surface **130** as the plunger **70** and actuator **20** are forced against each other. The difference in the material hardness of the crossbar **195** and actuator engagement section

85 preferably ensures that the crossbar **195** is recessed into the actuator engagement surface **130** when engaged, thereby preventing the actuator **20** from sliding over the plunger **70**. Because the plunger engagement section **175** prevents the actuator engagement section **85** from sliding either to the side or over the top of the plunger **70** when the actuator is turned and the contacts are welded, a technician is made aware that the contacts **65**, **67** are, in fact, welded together. If a technician applies increased force to attempt to open the switch **5**, a component will break or snap before the actuator **20** is forced into an “off position.” The user is preferably able to hear a part breaking and feel the difference before and after either or both the actuator **20** or plunger **70** breaks, alerting the technician to a failure of the switch **5**.

(41) While FIGS. **10-12** illustrate one embodiment of the invention, the actuator engagement section **85**, or actuator engagement surface **130**, and crossbar **195** may be shaped differently as long as the design allows the necessary torque so that the actuator **20** and plunger **70** are able to interlock. For example, as shown in FIGS. **14A** and **14B**, two additional embodiments of the actuator **20** and plunger **70** are illustrated. As illustrated in FIG. **14A**, the plunger **70** includes a crossbar **195** with an angled protrusion, while the actuator engagement surface **130** may be slightly slanted such that an upper portion **130A** of the actuator engagement surface **130**, or region closer to the actuator shoulder section **90**, is located closer to the plunger engagement sidewall **178** than a lower portion **130B** of the actuator engagement surface **130**. Turning to FIG. **14B**, the actuator engagement surface **130** may further be slanted or angled in such a way that the portion of the actuator **20** at the second actuator face **80** protrudes furthest toward the plunger and the actuator engagement surface **130** tapers inward as it extends away from the second actuator face **80**. The plunger engagement sidewall **178** may have a complementary slope such that a space between the actuator engagement surface **130** and the plunger engagement sidewall **178** remains generally constant. Thus, while the plunger **70** may not bite as far into the actuator **20**, the plunger engagement sidewall **178** more fully engages the actuator engagement surface **130** preventing the actuator **20** from sliding upward and over the plunger **70** if the contacts are welded when the actuator **20** is rotated.

(42) Moreover, while the examples illustrated in FIG. **14** show that the plunger **70** includes a plunger **195** at the top of the plunger engagement sidewall **178** and the sectional area of the plunger **195** is illustrated as an angled, or triangular, protrusion, the crossbar **195** may be any shape and size as long as it either bites into the actuator engagement surface **130** or engages a complementary shaped actuator engagement surface **130** in such a manner that the actuator **20** is not pushed upwards when additional torque is applied to the actuator **20**.

(43) In operation, the actuator handle **22** is rotated by a technician to selectively move the switch **5** between the first or “ON” state and the second or “OFF” state. As further illustrated in FIG. **15**, when the contacts **65**, **67** are free to move, rotation of the switch **5** in a first direction causes the cam **97** on the actuator **90** to engage the first gear tooth **141** on the internal actuator **147** pushing the plate **149** and, in turn, the plunger downwards to separate the moving contact **65** from the fixed contact **67**. Rotation of the switch in the opposite direction, causes the cam **97** on the actuator **90** to engage the second gear tooth **143** on the internal actuator, drawing the plate **149** and, in turn, the plunger upwards to close the moving contact **65** with the fixed contact **67**. In the “ON” state, the contacts **65**, **67** are closed, forming an electrical connection. In the “OFF” state, the contacts **65**, **67** the contacts are open, preventing current flow through the switch **5**.

(44) If however, the contacts **65**, **67** are welded together, rotation of the actuator **20** attempts to push the plate **149** and, in turn, the plunger **70** downwards, but the welded contacts prevent motion of the plunger. The actuator **20** and the plunger **70**, therefore, engage each other. The actuator engagement section **85** rotates towards the plunger engagement section **175** until the actuator engagement surface **130** is located within the engagement area **190** and abuts the crossbar **195**. If the actuator **20** continues to be rotate without the plunger **70** lowering, the torque applied to the actuator **20** causes the actuator engagement surface **130** to interlock with the crossbar **195**. The

harder material of the crossbar **195** engages the softer material of the actuator engagement surface **130**, thereby preventing upward motion of the actuator **20** and preventing the actuator **20** from slipping over the top edge of the plunger **70**. The actuator engagement section **85** is also located within the L-shaped engagement area **190** on the plunger, thereby preventing sideways motion of the actuator **20** and preventing the actuator **20** from slipping past the side of the plunger **70**. Because the actuator **20** is engaging the plunger **70**, if a technician applies further force in an attempt to rotate the actuator **20**, then either or both the actuator **20** and plunger **70** will break. The mechanism, therefore, does not allow the actuator **20** to be rotated to an “off” position when the contacts are welded together. When one, or both, the actuator **20** and plunger **70** breaks, the user preferably can hear the sound of the breaking components and feel the abrupt difference in resistance within the switch **5** as the actuator completes its rotation to the off position. The noise and tactile feedback provide an indication to the technician that the contacts are welded and, further, that either the actuator **20** or plunger **70** are now broken and the switch **5** needs repair or replacement.

(45) The novel structure of the actuator **20** and plunger **70** therefore allow users to quickly and safely determine whether contacts **65** have been opened under normal operation (i.e., an electrical connection has been successfully broken) without installing additional components or requiring additional expenses. The structure similarly either prevents rotation of the handle **22** to an OFF position or alerts the technician via the noise and tactile feedback of an internal component failure if the contacts are welded.

(46) It should be understood that the invention is not limited in its application to the details of construction and arrangements of the components set forth herein. The invention is capable of other embodiments and of being practiced or carried out in various ways. Variations and modifications of the foregoing are within the scope of the present invention. It also being understood that the invention disclosed and defined herein extends to all alternative combinations of two or more of the individual features mentioned or evident from the text and/or drawings. All of these different combinations constitute various alternative aspects of the present invention. The embodiments described herein explain the best modes known for practicing the invention and will enable others skilled in the art to utilize the invention.

(47) In the preceding specification, various embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

Claims

1. A switching device for industrial equipment having at least one pair of contacts, the switching device comprising: an actuator having an actuator engagement section, wherein the actuator is selectively rotated around an axis of rotation between a first position and a second position; and a plunger configured to move reciprocally along a plunger axis between a first position and a second position, wherein the plunger axis is parallel to the axis of rotation around which the actuator rotates, the plunger having: a first end with an L-shaped engagement area configured to engage the actuator engagement section, and a second end with at least one movable switching element mounted on the second end of the plunger, wherein the L-shaped engagement area includes: a first portion oriented transverse to a direction of rotation of the actuator, a second portion oriented generally orthogonal to the first portion and tangential to the direction of rotation of the actuator, and a crossbar protruding from the first portion and the second portion, wherein the crossbar protrudes from the first portion toward the actuator engagement section when the actuator is in the first position; wherein when the at least one pair of contacts are not welded together, the actuator

selectively moves the plunger along the plunger axis as the actuator is rotated between the first position and the second position, thereby moving the crossbar out of position to engage the actuator engagement section; and wherein when the at least one pair of contacts are welded together, the welded contacts prevent the plunger from moving along the plunger axis due to rotation of the actuator and the actuator engagement section engages the crossbar on the L-shaped engagement area of the plunger thereby preventing the actuator from rotating between the first position and the second position.

2. The switching device of claim 1 wherein the actuator further includes a cam configured to engage an internal drive member, wherein: the internal drive member is selectively moved between a first position and a second position by the cam as the actuator is selectively moved, and the internal drive member is coupled to the plunger to selectively move the plunger along the plunger axis between the first position and the second position as the internal drive member is selectively moved.

3. The switching device of claim 1, wherein the crossbar is configured to compress into a surface of the actuator engagement section when the contacts are welded and the actuator is moved from an “ON” position to an “OFF” position.

4. The switching device of claim 1 wherein at least a portion of the plunger or at least a portion of the actuator breaks when the at least one pair of contacts are welded together and the actuator is moved between the first position and the second position.

5. The switching device of claim 4 wherein breaking either the actuator or plunger provides at least one of a tactile and an audible alert that the contacts are welded.

6. The switching device of claim 1 wherein the actuator engagement section is formed from a first material and the plunger engagement section is formed from a second material, wherein the second material is harder than the first material.

7. The switching device of claim 6 wherein the first material is selected from at least one of nylon, polyoxymethylene, and polybutylene terephthalate.

8. The switching device of claim 6 wherein the second material is selected from one of polyether ether ketone reinforced with carbon fiber, glass reinforced polyphenylene sulfide, polyamides with glass particles, and nylon with glass fiber.
